

3.6 Anisotropic earth structure

3.6.1 General considerations

So far in this chapter, we have considered a view of the earth developed from analyses of seismic waves assuming that they propagated through an earth made up of purely isotropic, linearly elastic material (Section 2.3.9). In such material, the stresses are linearly proportional to the strains via Hooke's law

$$\sigma_{ij} = c_{ijkl} \epsilon_{kl}, \quad (1)$$

and the 81-term tensor of elastic moduli, c_{ijkl} , reduces to two independent elastic constants, λ and μ . As a result, the material's elastic properties are the same in all directions. Although isotropy is a good first approximation in the earth, it is sometimes important to consider deviations from isotropy, or *anisotropy*. In such cases, Hooke's law applies, but the relation between the stresses and strains involves more than two elastic constants. Although there can be up to 21 independent elastic constants, any material in which more than two are needed is called *anisotropic*.

Having more than two elastic constants means that the material's properties differ depending on the direction. Because seismic wave velocities depend on the elastic constants, waves traveling through anisotropic material travel faster or slower depending on their direction, and complicated wave phenomena can occur. For example, a shear wave can be split into two pulses, each with a different polarity and traveling at a different speed (Figs 3.6-1, 2.4-8).

Anisotropy can result from a material's being non-uniform, a condition called *heterogeneity* or *inhomogeneity*. A common situation is when material has directionality in its structure. For instance, plywood is a superposition of thin layers of wood, so its strength (shear modulus) differs in different directions.

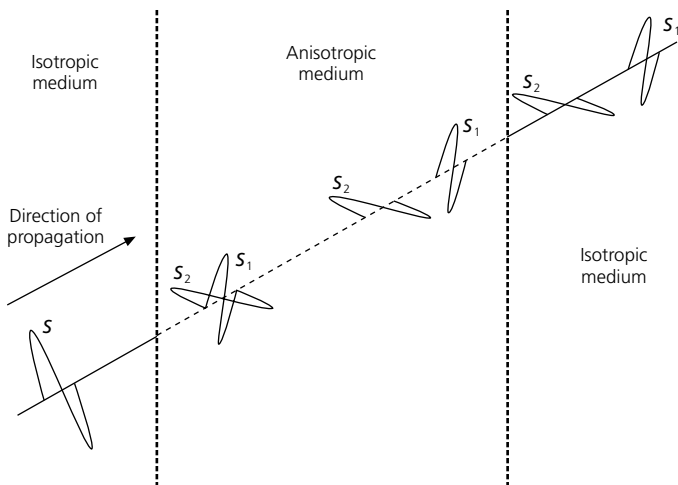


Fig. 3.6-1 Schematic of an initially polarized shear wave split along the fast and slow anisotropic directions, yielding pulses separated in time. The pulses remain split after leaving the anisotropic region.

Similarly, a stack of rock layers with different isotropic velocities can as a whole behave anisotropically, so seismic waves travel with different speeds parallel or perpendicular to the layers. This situation is called shape-preferred orientation (SPO) anisotropy. Anisotropy can also occur for homogeneous materials. For example, the crystal structure of the mineral olivine is homogeneous in that it is composed of the same repeating groups of atoms, but acts anisotropically because its acoustic properties vary in different directions relative to the crystal lattices of an agglomeration of mineral grains. This situation is called lattice-preferred orientation (LPO) anisotropy.

The anisotropic variations of the seismic velocity of earth materials are small compared to the large changes in seismic velocity that occur radially from the surface to the core. Hence, in developing radial models of seismic velocity, anisotropy has traditionally been treated as a secondary effect. Nevertheless, recent efforts to better quantify three-dimensional velocity variations sometimes find that anisotropic perturbations are comparable to lateral velocity changes. It is often difficult, however, to distinguish between the effects of anisotropy and those of heterogeneity. For example, curvature on a refracting interface can simulate many of the effects associated with anisotropy.

An important reason to study anisotropy is that material flow at depth appears to preferentially orient olivine crystals within upper mantle rocks. Hence mapping the seismically "fast" direction lets us investigate the relation between plate motions and mantle flow at depth. Although anisotropy studies are ongoing, and both results and interpretations will change over time, they represent a major frontier in deep earth studies.

3.6.2 Transverse isotropy and azimuthal anisotropy

As discussed in Section 2.3.9, the symmetry of the stress and strain tensors and the idea of strain energy means that no more than 21 of the 81 elastic constants c_{ijkl} are independent. We can thus write the c_{ijkl} tensor as a matrix C_{mn} , where the indices m and n vary from 1 to 6 as the pairs of indices (i, j) or (k, l) take values of (1, 1), (2, 2), (3, 3), (2, 3), (1, 3) and (1, 2), respectively:

$$C_{mn} = \begin{pmatrix} c_{1111} & c_{1122} & c_{1133} & c_{1123} & c_{1113} & c_{1112} \\ c_{2211} & c_{2222} & c_{2233} & c_{2223} & c_{2213} & c_{2212} \\ c_{3311} & c_{3322} & c_{3333} & c_{3323} & c_{3313} & c_{3312} \\ c_{2311} & c_{2322} & c_{2333} & c_{2323} & c_{2313} & c_{2312} \\ c_{1311} & c_{1322} & c_{1333} & c_{1323} & c_{1313} & c_{1312} \\ c_{1211} & c_{1222} & c_{1233} & c_{1223} & c_{1213} & c_{1212} \end{pmatrix} = \begin{pmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ C_{21} & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ C_{31} & C_{32} & C_{33} & C_{34} & C_{35} & C_{36} \\ C_{41} & C_{42} & C_{43} & C_{44} & C_{45} & C_{46} \\ C_{51} & C_{52} & C_{53} & C_{54} & C_{55} & C_{56} \\ C_{61} & C_{62} & C_{63} & C_{64} & C_{65} & C_{66} \end{pmatrix}. \quad (2)$$

For an isotropic material, the c_{ijkl} tensor can be written in terms of two independent elastic constants

$$c_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad (3)$$

so its matrix form is

$$C_{mn} = \begin{pmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{pmatrix}. \quad (4)$$

However, the crystal structures of many earth materials require additional independent elastic coefficients. For example, ice, quartz, olivine, or plagioclase feldspar require 5, 6, 9, and 21 constants, respectively. In such cases, the matrix is more complicated.

One of the most important forms of anisotropy, known as *transverse isotropy* (also known as radial anisotropy, axisymmetry, and cylindrical symmetry), occurs for a stack of layered materials. Each layer is isotropic in its properties, but these properties differ between layers (as in plywood). Thus the elastic properties, and hence seismic velocities, of the stack as a whole are identical regardless of the amount of rotation about the axis of symmetry, which is perpendicular to the layers. However, these aggregate properties differ in the perpendicular directions.

A transversely isotropic material can be characterized by five independent elastic coefficients, A, C, F, L, N , that represent its aggregate properties. If the axis of symmetry is x_3 , so properties in that direction differ from those in the x_1 - x_2 plane, the elastic constant matrix (Eqn 4) becomes

$$C_{mn} = \begin{pmatrix} A & A - 2N & F & 0 & 0 & 0 \\ A - 2N & A & F & 0 & 0 & 0 \\ F & F & C & 0 & 0 & 0 \\ 0 & 0 & 0 & L & 0 & 0 \\ 0 & 0 & 0 & 0 & L & 0 \\ 0 & 0 & 0 & 0 & 0 & N \end{pmatrix}. \quad (5)$$

Comparisons with matrices 2 and 4 show that terms that were the same for an isotropic material (consider C_{11} and C_{33} , or C_{55} and C_{66}) now differ, because terms involving the x_3 direction differ from those in the x_1 or x_2 directions.

This matrix gives the velocities of waves propagating in different directions. First, consider waves propagating in the x_1 direction (Fig. 3.6-2, *top*). By analogy to the isotropic case, A corresponds to $\lambda + 2\mu$ for the x_1 direction, N corresponds to μ for the x_2 direction, and L corresponds to μ for the x_3 direction. Thus the P velocity and the two orthogonal S velocities are

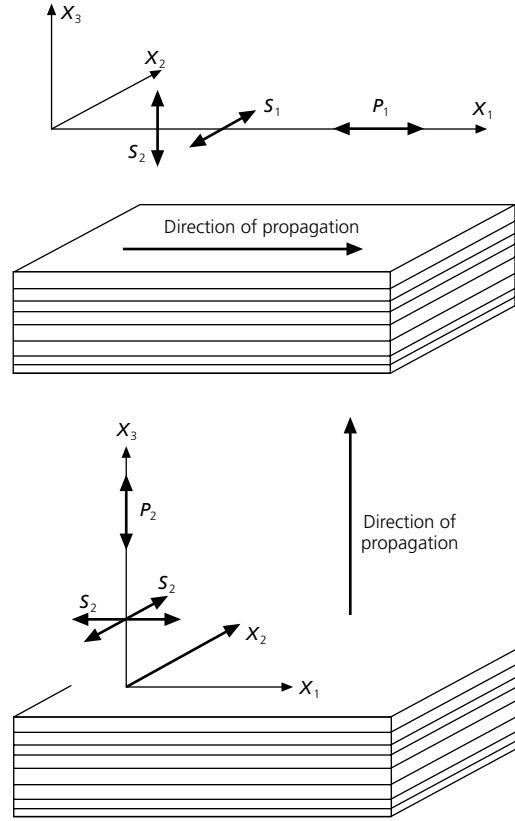


Fig. 3.6-2 Cartoon showing the effects of transverse isotropy due to layering. *Top*: Directions of oscillations for P and S waves propagating in the x_1 direction, in the plane of layering. The shear wave oscillating in the plane of the layering has velocity S_1 , which is generally faster than that for the shear wave oscillating across the layers, S_2 . *Bottom*: Directions of oscillations for P and S waves propagating in the x_3 direction, perpendicular to the layering. The compressional wave velocity, P_2 , is generally less than P_1 . Both shear waves have the same velocity, S_1 .

$$P_1 = (A/\rho)^{1/2}, \quad S_1 = (N/\rho)^{1/2}, \quad S_2 = (L/\rho)^{1/2}. \quad (6)$$

Hence the velocity of shear waves traveling in this direction depends on the directions of their particle motions. The waves become split, with waves polarized in one plane traveling faster than those polarized in the other. This is one way to get splitting like that shown in Fig. 3.6-1. These results would be the same for propagation in the x_2 direction, or any other direction in the x_1 - x_2 plane, because physical properties in this plane are independent of direction.

In many applications, the horizontally layered earth shows transverse isotropy about a vertical axis. The SH -wave velocity S_1 is generally faster than the SV velocity S_2 , because the SH displacement is preferentially in the fast layers, whereas SV samples both equally. An interesting consequence is that the shear velocity inferred from the dispersion of Love waves, which are SH waves, would be higher than that from Rayleigh waves, which involve SV .

By contrast, for P and S waves propagating in the x_3 (axis of symmetry) direction (Fig. 3.6-2, *bottom*), both S velocities equal S_2 in Eqn 6. The P velocity reflects the fact that C corresponds to $\lambda + 2\mu$ for the x_3 direction, so

$$P_2 = (C/\rho)^{1/2}. \quad (7)$$

For layered materials, typically $P_1 > P_2$, so P waves propagate faster in the x_1 direction than in the x_3 direction. This is because the wave travels preferentially in the fast layers in the x_1 direction, whereas a P wave traveling in the x_3 direction must also traverse the slow layers.

Transverse isotropy is often characterized by three parameters:

$$\xi = N/L = (S_1/S_2)^2, \quad \phi = C/A = (P_2/P_1)^2, \quad \eta = F/(A - 2L). \quad (8)$$

If the material were isotropic, $\xi = \phi = \eta = 1$. For layered structures, generally $\xi > 1$ and $\phi < 1$.

A second common type of anisotropy is *azimuthal anisotropy*, in which velocities vary as a function of horizontal direction. One way to obtain this is to have transverse isotropy with the x_3 axis turned to horizontal, which is analogous to standing plywood vertically. In general, the P -wave velocity varies with azimuth as

$$P(\theta) = A_1 + A_2 \cos 2\theta + A_3 \sin 2\theta + A_4 \cos 4\theta + A_5 \sin 4\theta, \quad (9)$$

where the constants A_i depend on the 21 elastic constants.

3.6.3 Anisotropy of minerals and rocks

An important source of seismic anisotropy is minerals that are anisotropic due to their crystal structure. At microscopic levels the anisotropy can be enormous, with velocities along different mineralogical axes varying by more than 100%. Generally, however, the anisotropic mineral grains are randomly oriented, so seismic waves have wavelengths long enough to average out the anisotropic effects, leaving only weak anisotropy. However, in some cases the mineral grains are aligned, causing significant anisotropy.

Laboratory studies of the elastic moduli of minerals give insight into such LPO anisotropy. Some studies involve static methods like twisting or squeezing samples, but most use the vibrational properties of mineral samples as small as 1 mm. At very high pressures, a technique called Brillouin scattering, which measures how laser light passing through the mineral is distorted, yields elastic constants for samples smaller than 0.1 mm.

One of the most important anisotropic minerals is olivine (Fig. 3.6-3), which comprises much of the upper mantle (Section 3.8). For waves propagating in the fastest direction, the P -wave velocity is 9.89 km/sec and the S velocities are 4.89 km/s and 4.87 km/s. By contrast, the slowest P velocity in

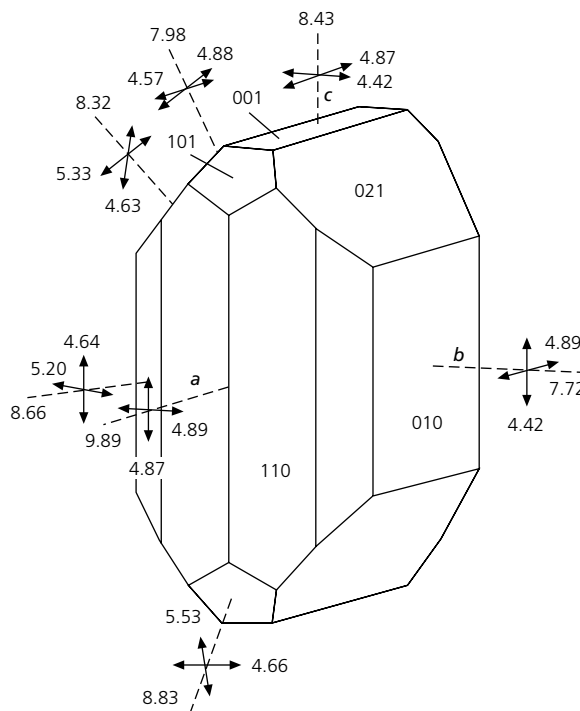


Fig. 3.6-3 P and S velocities (km/s) in different directions relative to the crystal structure of olivine. P velocities are in the directions of the dashed lines, and the S velocities are shown by the adjacent pairs of perpendicular lines. The a axis, corresponding to the $[100]$ crystal face, is the fastest direction through the crystal. It is also the dominant slip direction, so olivine crystals align in the direction of plastic flow. (Babuska and Cara, 1991. With kind permission from Kluwer Academic Publishers.)

this example is 7.72 km/s. The magnitude of anisotropy is characterized by

$$k = (v_{\max} - v_{\min})/v_{\text{mean}}. \quad (10)$$

For P -waves in the olivine crystal, $\alpha_{\max} = 9.89$ km/s, $\alpha_{\min} = 7.72$ km/s, and $\alpha_{\text{mean}} = 8.81$ km/s, so $k = 25\%$. The maximum and minimum S velocities are 5.53 km/s and 4.42 km/s, so $k = 22\%$. Although for olivine the anisotropy of P and S waves is similar, they can differ greatly for other minerals.

Other important minerals range from nearly isotropic to extremely anisotropic. One of the most isotropic minerals is garnet, where k for both P and S waves is $\leq 1\%$. At the other extreme, sheet silicates like mica can have values of k up to 60% for P waves and 116% for S waves.

As a result, a major factor controlling a rock's anisotropy is the anisotropy of the minerals composing it and their relative proportions. Another important factor is the presence of deviatoric stresses, which can cause a preferred orientation of anisotropic mineral grains that might otherwise be randomly distributed. Crystals are generally oriented with their smallest widths in the direction of maximum compression. For example,

in micas, which are important components of highly foliated schists, the flat crystals are oriented parallel to the plane of least compression. Thus slip occurs more easily parallel to the developing foliation, because the planar mica faces contain the weakest bonds. Shear in a preferred direction can also recrystallize different mineral assemblages, so the resulting anisotropy reflects a combination of the preferred orientation of anisotropic materials and the presence of laminar structures.

3.6.4 Anisotropy of composite structures

Anisotropy can also result from an asymmetric combination of materials. The upper continental crust often contains horizontally layered sedimentary rocks. Similarly, oceanic crust is comprised of sediments overlying layers of basalt and gabbro. Such layering can yield transverse isotropy, with the symmetry axis oriented vertically. On a regional scale, plate collisions often cause significant metamorphism, sometimes yielding transverse isotropy due to the preferred orientation of the foliation of gneisses and schists.

Fluid-filled cracks, for example in a volcanic region, can also cause anisotropy. For a material containing two-dimensional fluid-filled cracks whose normals are parallel to the x_1 axis, the anisotropy is given by

$$C_{ij} = \begin{pmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu(1 - \varepsilon) & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu(1 - \varepsilon) \end{pmatrix}, \quad (11)$$

where ε is the crack density given by $\varepsilon = Na^3/V$, N is the number of cracks in the volume V , and a is the half-width of a crack. If the cracks become infinitely small, $\varepsilon = 0$, yielding the isotropic case (Eqn 4). In general, the anisotropy depends on the geometry of the inclusions and their contrast in properties with the surrounding matrix. For computational ease, rods (prolate spheroids) and disks (oblate spheroids) are often assumed in seismic modeling.

3.6.5 Anisotropy in the lithosphere and the asthenosphere

Anisotropy in the lithosphere takes many forms, including that in glaciers whose flow aligns the ice crystals. Closer to our applications, several effects generate anisotropy in the oceanic crust. Horizontal sediment layers can create transverse isotropy of up to 15% with a vertical symmetry axis. In the upper crustal layer of vertical-sheeted basaltic dikes, azimuthal anisotropy is thought to exist with a horizontal axis perpendicular to the dikes and thus in the spreading direction.

Sub-crustal oceanic lithosphere shows strong azimuthal anisotropy. The flow processes associated with plate spreading

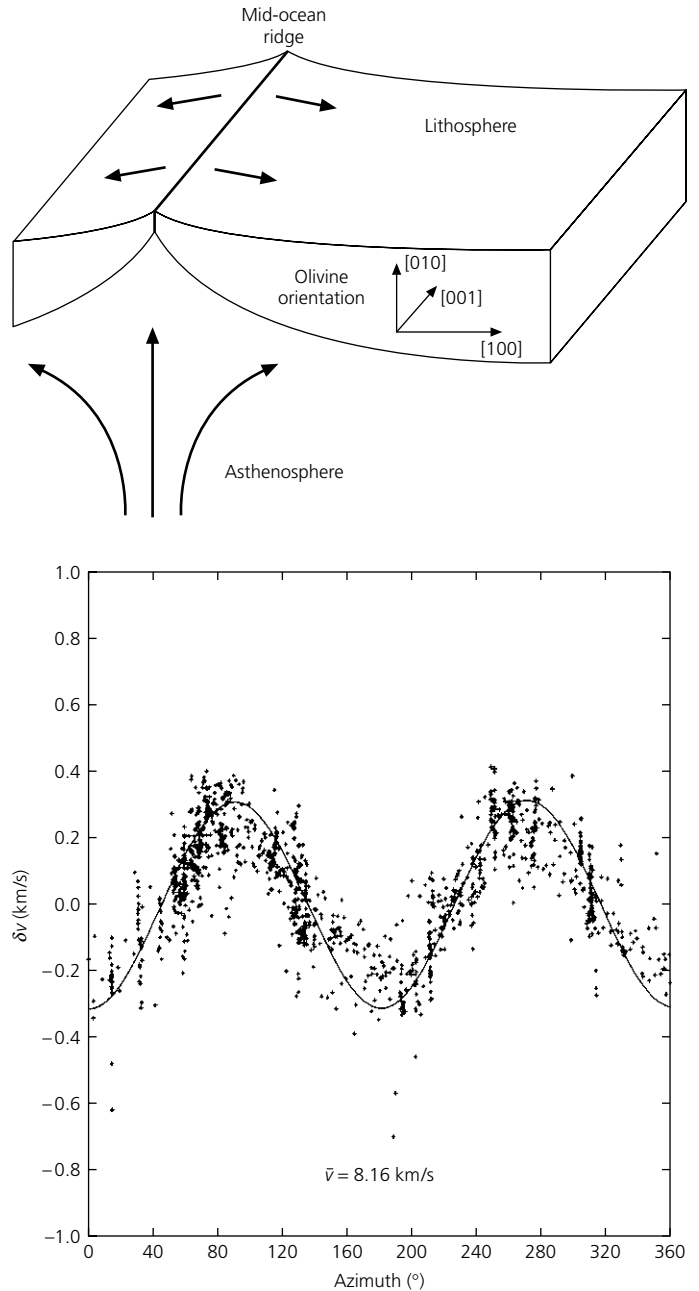
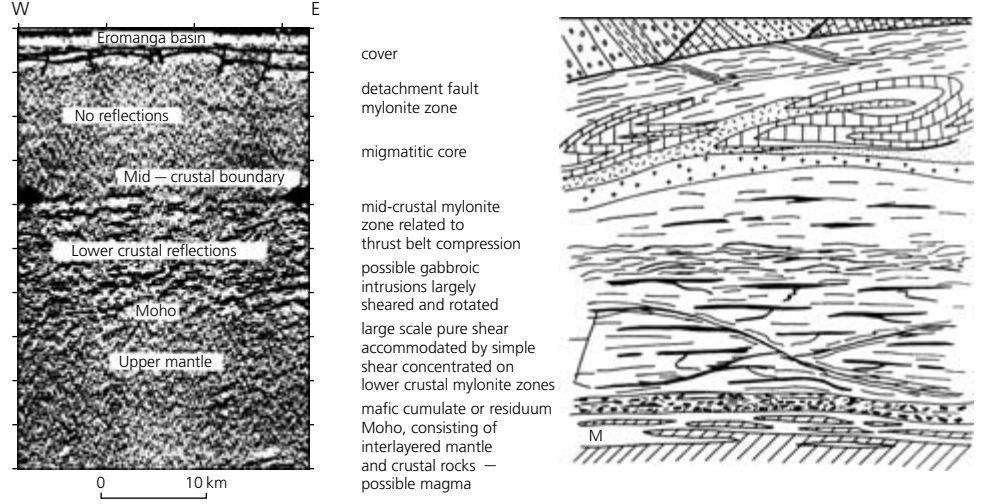


Fig. 3.6-4 *Top:* Illustration of how the spreading process yields a preferred orientation of olivine crystals in the oceanic lithosphere, with the fast axis of velocity ([100]) in the spreading direction. *Bottom:* Variations in P_n wave velocities near Hawaii. The azimuth is measured relative to the trend of the isochrons (90° from the spreading direction), so the maxima at 90° and 270° show that the fast direction of the azimuthal anisotropy is in the direction of spreading when the plate formed. (Morris *et al.*, 1969. *J. Geophys. Res.*, 74, 4300–16, copyright by the American Geophysical Union.)

Fig. 3.6-5 *Left*: Seismic reflection profile of the crust and upper mantle in eastern Australia. The lower crust has multiple, discontinuous, and sub-horizontal reflectors possibly due to strain-induced fabrics, igneous layering, or free fluids. This structure yields vertical-axis transverse isotropy. (Finlayson *et al.*, 1989. *Properties and Processes of Earth's Lower Crust*, 1–16, by permission of Australian Geological Survey Organisation.) *Right*: Schematic cross-section of the crust in the northern Ruby Mountains of the North American Basin and Range. There is a strong tendency toward horizontally layered features, although the likely origins of such fabrics vary with depth. (Smithson, 1989. *Properties and Processes of Earth's Lower Crust*, 53–63, permission as above.)



(Fig. 3.6-4, *top*) appear to orient olivine crystals preferentially in the spreading direction, along their [100] slip axes.¹ Because P waves propagate fastest in this direction (Fig. 3.6-3), P_n head waves that sample the uppermost mantle just below the Moho (Section 3.2.1) show a strong azimuthal velocity dependence (Fig. 3.6-4, *bottom*). This variation is approximately described by the $\cos 2\theta$ term in Eqn 9, where θ is measured from the spreading direction, so the velocity is highest in the spreading direction or 180° from it. This anisotropy is “frozen in” as the lithosphere ages, and so records the spreading direction.

Because continental crust is more complicated than oceanic crust, so is its anisotropy. A primary source of anisotropy in the upper crust is the presence of cracks, often fluid-filled. Such cracks often have a near-vertical orientation induced by regional stress fields parallel to the cracks. When these cracks occur in horizontal sediments that would by themselves have vertical-axis transverse isotropy, the combined result can be orthorhombic symmetry. The lower continental crust tends to have strong sub-horizontal layering, perhaps resulting from ductile deformation, which causes seismic anisotropy. Figure 3.6-5 shows such layering in a seismic reflection profile and a schematic diagram.

Anisotropy within and beneath continental lithosphere is often studied with a technique called *shear wave splitting*. When SKS waves convert from P waves in the outer core to S waves in the lower mantle, they are entirely polarized in the radial (SV) direction, because all the initial SH energy was reflected when the downgoing S wave encountered the core–mantle boundary. As these shear waves travel across the mantle and crust, however, they can be *split* when traveling through anisotropic media (Fig. 3.6-6). Assuming transverse isotropy with a horizontal axis of symmetry, the two polarized waves travel at different

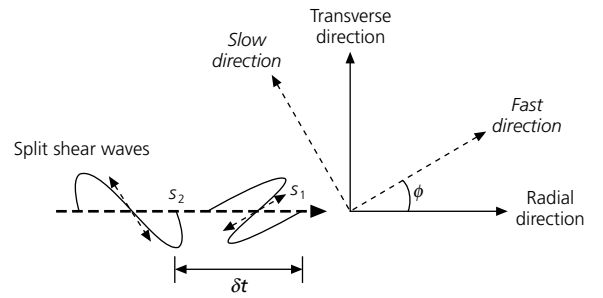


Fig 3.6-6 Splitting of an incoming shear wave into pulses oriented along the fast (s_1) and slow (s_2) directions of anisotropy. The polarization angle ϕ gives the rotation of the fast axis relative to the radial propagation direction, and δt is the time difference between the split pulses.

speeds and arrive at different times. Thus, if the SKS signal on the radial component in an isotropic earth is $s(t)$, its projection into the fast and slow polarizations is, respectively,

$$s_1(t) = s(t) \cos \phi, \quad -s_2(t) = s(t - \delta t) \sin \phi, \quad (12)$$

where ϕ is the polarization angle between the radial direction and the fast axis, and δt is the delay time between the fast and slow polarizations. We would normally not expect any SKS on the transverse component, but anisotropy yields a combination of both the fast and the slow polarizations on both the radial and the transverse components, given by

$$R(t) = s(t) \cos^2 \phi + s(t - \delta t) \sin^2 \phi, \\ T(t) = [(s(t) - s(t - \delta t))/2] \sin 2\phi. \quad (13)$$

For example, in Fig. 3.6-7a (*top*), SKS appears on the transverse component. The two components are rotated to yield the fast and slow polarizations, $s_1(t)$ and $s_2(t)$ (Fig. 3.6-7a, *middle*). The time shift δt is then applied, and the signals are rotated

¹ This representation of crystallographic axes is discussed in mineralogy texts like Klein and Hurlbut (1985).

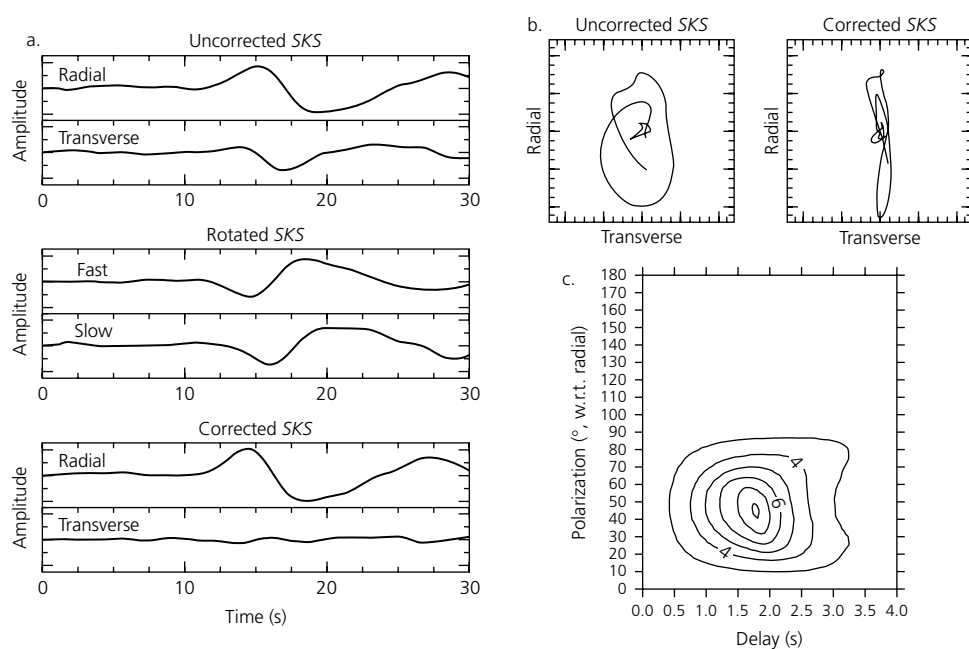


Fig. 3.6-7 Shear wave splitting of SKS waves for a Kuril Islands earthquake, stacked across an array of seismometers in New Zealand. *a*: SKS waveforms before and after processing. *Top*: radial and transverse components before processing. Note the large SKS signal on the transverse component, which should not be there for an isotropic earth. *Middle*: SKS waveforms after rotation into the fast and slow polarizations. *Bottom*: SKS waveform after the splitting has been removed so that all SKS is on the radial component. *b*: Particle motion plots (Section 2.4) of SKS on the radial and transverse components before and after removal of the transverse signal. *c*: Contour plot of the amplitude in the radial component as a function of the delay time and polarization angle. The minimum corresponds to the best-fitting value. (Gledhill and Gubbins, 1996, *Phys. Earth Planet. Inter.*, 95, 227–36, with permission from Elsevier Science.)

again so that all of the signal appears on the radial component (Fig. 3.6-7a, *bottom*). As shown in Fig. 3.6-7b, before correction, particle motion occurs on both components, but after correction, the motion is limited to the radial component. The fact that this technique removes the transverse signal shows the appropriateness of the transversely isotropic model. The values of ϕ and δt are found by minimizing the transverse signal, as shown by the contour plot in Fig. 3.6-7c. Typical values for the magnitude of shear wave splitting, δt , are in the 0–2 s range.

Seismic anisotropy within continents is thought to reflect crystal alignment created during a tectonic episode and then “frozen in.” The anisotropy is a result of the last episode of tectonism, which resets any previous anisotropy. Because continental rock can be as old as 4 Ga (the mean age is about 1.5 Ga), anisotropy in continental lithosphere can reveal information about very old tectonic events such as episodes of mountain building. For plate collisions the fast axis is usually sub-perpendicular to the principal stress axis, or parallel to the resulting orogenic belts. There may also be deeper anisotropy due to oriented olivine in the flowing asthenosphere. However, it is sometimes difficult to distinguish this effect from lithospheric anisotropy. For instance, in eastern North America the fast axis is oriented WSW–ENE, parallel to the direction of both absolute plate motion (Section 5.2.4) (and thus presumably asthenospheric flow) and major orogenic boundaries like the Appalachian Mountains (Fig. 3.6-8).

Surface wave observations indicate that anisotropy extends to a depth of about 300 km beneath oceans. The S -wave velocity inferred from Love waves, which are SH waves, is higher than inferred from Rayleigh waves, which involve SV . Figure 3.6-9 shows the squared S -wave velocity ratio ξ (Eqn 8) versus

depth for several ages of oceanic lithosphere. The deviation of ξ from 1 reflects transverse isotropy with SH velocities faster than SV velocities. Because the oceanic lithosphere extends to a depth of about 100–125 km, anisotropy seems to extend into the asthenosphere.

In addition, Rayleigh wave velocities show azimuthal anisotropy similar to that found for P_n waves that sample the uppermost mantle at much shallower depths. Both types of anisotropy may reflect mantle flow (Fig. 3.6-4). The flow-induced preferred orientation of olivine would give azimuthal anisotropy in the spreading direction. Taking paths in different directions averages out the azimuthal effect, leaving a net transverse isotropy that is symmetric about the vertical. An interesting consequence of this model is that near the ridges, where mantle material is upwelling, transverse isotropy should be less significant, as the data show. At older ages, mantle flow will be more horizontal, increasing transverse isotropy.

3.6.6 Anisotropy in the mantle and the core

Although most of the mantle shows little or no anisotropy, this is not so for the D'' region at the base of the mantle, where complex interactions with the liquid outer core may occur (Section 3.5.4). Studying anisotropy in a narrow layer nearly 3000 km below the heterogeneous mantle and crust is challenging, but initial investigations suggest anisotropy on the order of several percent, comparable to the isotropic velocity variations. D'' anisotropy seems to fall in to two categories. Beneath regions of paleo-subduction, such as western Central America and the northern Pacific rim, SH waves in the form of S , ScS , or S_{diff} travel faster than their SV counterparts (Fig. 3.6-10). This behavior has been modeled as transverse isotropy. However,

Fig. 3.6-8 Map of the eastern USA showing shear wave splitting results from SKS and SKKS. Lines point in the direction of the fast axis, assuming horizontally oriented transverse isotropy, and the sizes of the circles represent the magnitude of the splitting in seconds. The background is a map of the shear wave velocity anomalies at 200 km depth (van der Lee and Nolet, 1997. *J. Geophys. Res.*, 102, 22, 815–38, copyright by the American Geophysical Union.) The splitting direction is approximately parallel to the Appalachian orogenic belts (dashed line) and aligned with the absolute plate motion (APM). Note the regional variations for different locations. (Fouch *et al.*, 2000. *J. Geophys. Res.*, 105, 6255–76, copyright by the American Geophysical Union.)

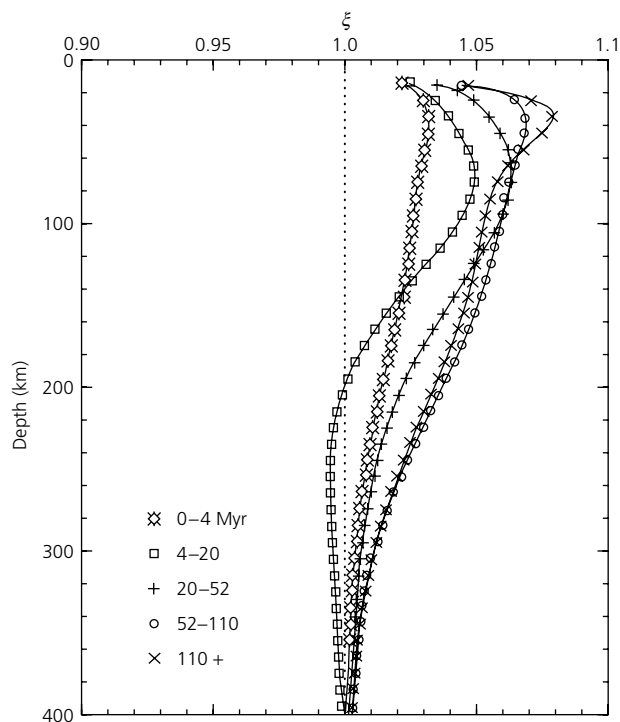
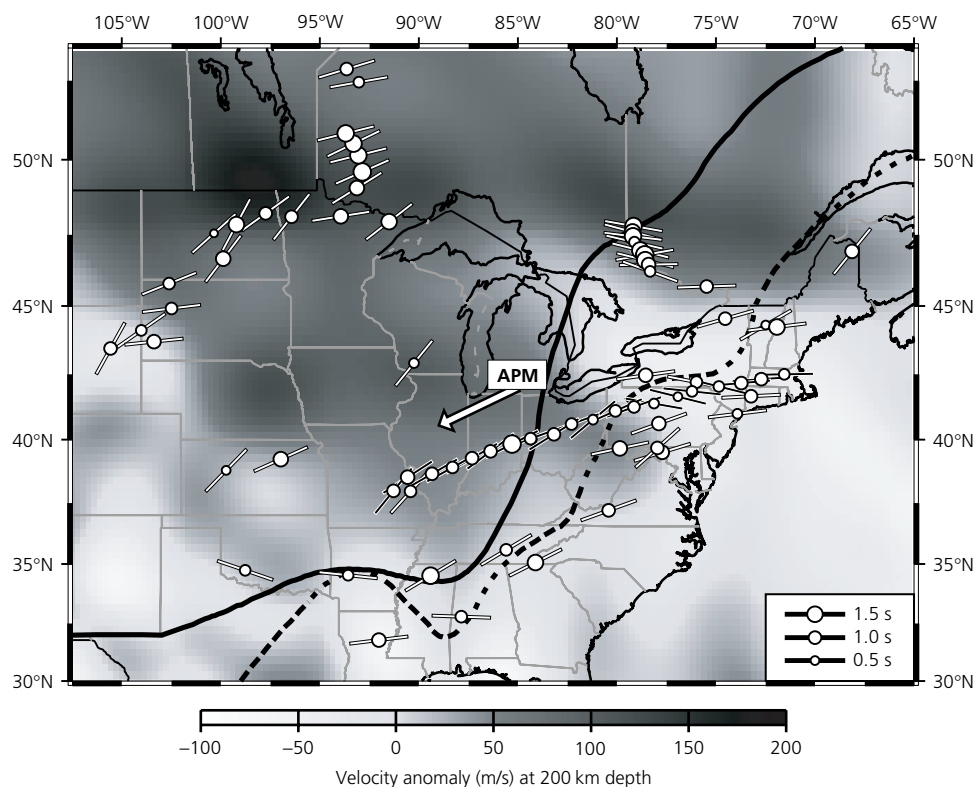


Fig. 3.6-9 Depth variations of ξ , the square of the V_{SH}/V_{SV} ratio, beneath the Pacific Ocean. ξ tends to exceed 1, meaning that SH is faster than SV , consistent with olivine in both the lithosphere and the asthenosphere being preferentially oriented by the spreading process. (Nishimura and Forsyth, 1989.)

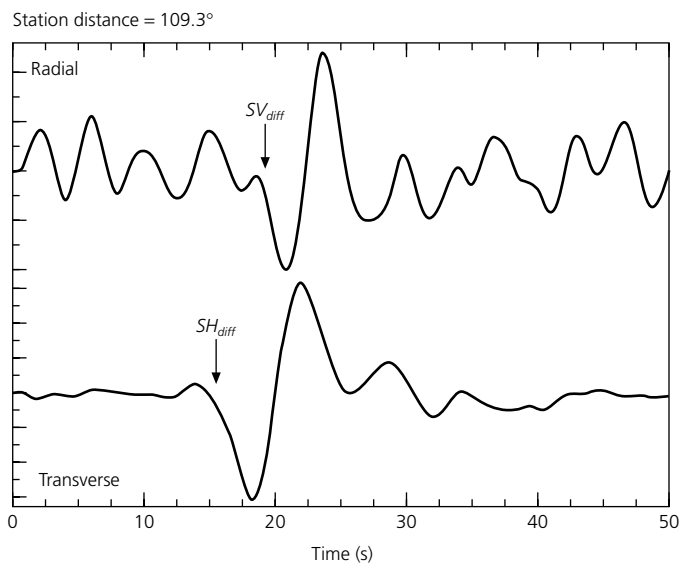


Fig. 3.6-10 Evidence for anisotropy at the base of the mantle, shown by diffracted arrivals for a South American earthquake recorded at Canadian station DAWY. Arrows show estimates of the onset times. Diffracted SH arrives before SV , suggesting transverse isotropy. (Kendall and Silver, 1996. Reproduced with permission from *Nature*.)

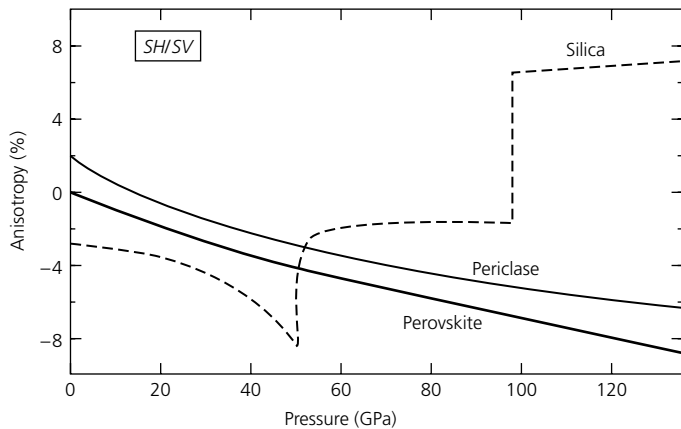


Fig. 3.6-11 Predicted anisotropic behavior for perovskite, periclase, and silica as a function of pressure in the mantle. The far right corresponds to the lowermost mantle, where these phases are major components. The kinks in the silica curve result from phase transitions. (Stixrude, 1998. *The Core–Mantle Boundary Region*, 83–96, copyright by the American Geophysical Union.)

D'' anisotropy beneath the mid-Pacific is variable, with SH waves usually but not always arriving before the accompanying SV waves. This effect may reflect vertical structures due to lower-most mantle upwelling. In addition, several mineral phases that are expected here, such as perovskite (MgSiO_3), periclase (MgO), and the columbite phase of silica (SiO_2), should be anisotropic under these conditions (Fig. 3.6-11). Because little of the core–mantle boundary has been examined for anisotropy due to the stringent earthquake-station geometries required, much is yet to be learned.

Significant anisotropy occurs in the solid iron inner core. PKIKP waves (PKP-DF) travel ~ 3 s faster in the inner core along the earth's rotation axis than along the equatorial plane. The PKP-DF and PKP-BC phases (Fig. 3.5-7) travel similar paths through the mantle, so any travel time difference between them is likely to reflect structure in the core. Because of the low viscosity of the liquid outer core, flow should eliminate any lateral velocity variations, including anisotropy. Thus the difference between the observed differential travel times of the BC and DF phases and that predicted by a model

$$\delta t_{BC} - \delta t_{DF} = (t_{BC} - t_{DF})_{\text{observed}} - (t_{BC} - t_{DF})_{\text{predicted}}, \quad (14)$$

is likely to be a function of inner core structure along the DF path.

Figure 3.6-12 shows BC-DF residuals versus ξ , the angle between the PKP-DF ray segment in the inner core and the earth's spin axis. Small values of ξ correspond to paths parallel to the spin axis, and the corresponding large residuals indicate that near-axial PKP-DF waves travel faster and arrive sooner. Also shown are theoretical predictions for the anisotropic behavior of solid iron in the hexagonal close-packed (hcp) and

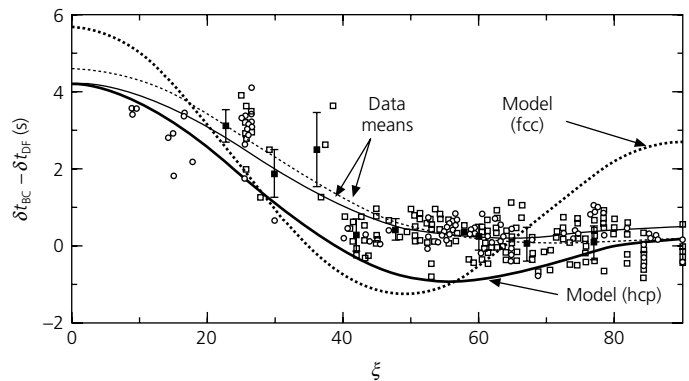


Fig. 3.6-12 PKP-BC – PKP-DF travel time residuals as a function of ξ , the angle between the PKP-DF ray and the earth's spin axis. Circles and thin solid line are for data from Song and Helmberger (1993); squares and thin dashed line are for data from Creager (1992). The thin solid and dashed lines are the smoothed fits to the residuals. The heavy solid and dashed lines are the predicted residuals for the transverse isotropy expected if the inner core were composed of iron in either the hcp and the fcc structures. The similarity between the hcp curve and the data support hcp as the crystal phase for the inner core. (Stixrude and Cohen, 1995. *Science*, 267, 1972–5, copyright 1995 American Association for the Advancement of Science.)

face-centered-cubic (fcc) structures. The hcp structure of iron, aligned along the earth's rotation axis, does a good job of modeling the observations.

Inner core anisotropy is also shown by normal modes that have significant displacement in the inner core. If there were no lateral heterogeneity or anisotropy, the various singlets making up a normal mode multiplet would have almost identical eigenfrequencies (Section 2.9). In fact, as shown in Fig. 3.6-13 for the $_{18}S_4$ multiplet, the modes are split, so the eigenfrequencies for the different singlets (points) vary depending on the azimuthal order. The solid line (left) shows the splitting predicted from a transversely isotropic model with elastic parameters (shown on the right). Here α , β , and γ are combinations of the elastic constants for transverse isotropy (Eqn 5). The velocity perturbation for any direction through the inner core is

$$\delta v/v = (2\beta - \gamma) \cos^2 \xi, \quad (15)$$

where ξ is the angle between the ray path and the earth's rotation axis. $\delta v/v$ is zero along an equatorial path, but is about 1% parallel to the axis.

Inner core anisotropy is not perfectly symmetric about the rotation axis, which allows for the possibility of observing differential rotation of the inner core with respect to the mantle. This phenomenon has been reported, seen as temporal variations of the BC-DF residuals (Eqn 14) for similar earthquake-station geometries. Quantification of such differential rotation and its implications for the generation of the magnetic field in the convecting outer core are active research areas.

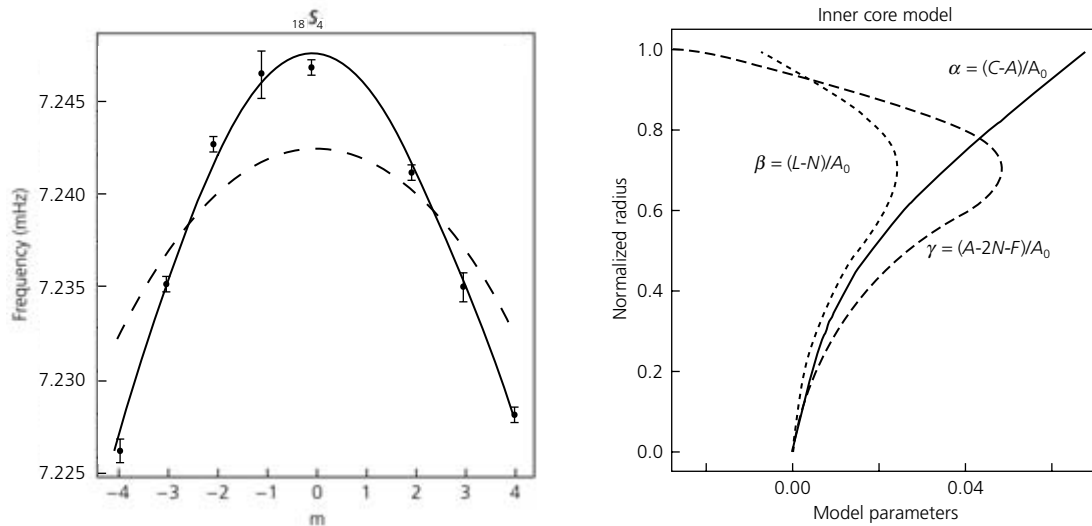


Fig. 3.6-13 Evidence for transverse isotropy in the inner core from the splitting of a spheroidal normal mode multiplet. Points represent the observed frequencies for the different azimuthal orders (*left*). The dashed curve is the prediction for the inner core including the effects of the earth's rotation and ellipticity but not anisotropy. The solid curve incorporates the anelastic model at the *right* by combinations of the elastic constants for transverse isotropy. In this formulation, α , β , and γ differ from their usual seismological definitions. (Tromp, 1993. Reproduced with permission from *Nature*.)

3.7 Attenuation and anelasticity

3.7.1 Wave attenuation

In the last section, we extended our view of the earth as an isotropic elastic medium to include the effects of anisotropy. We now consider *anelasticity*, or deviation from elasticity, which is one of the reasons why seismic waves *attenuate* or decrease in amplitude as they propagate. We have already discussed how the reflection and transmission of seismic waves at discrete interfaces reduce their amplitudes. Here, we consider four other processes that can reduce wave amplitudes: *geometric spreading*, *scattering*, *multipathing*, and anelasticity. The first three are elastic processes, in which the energy in the propagating wave field is conserved. By contrast, anelasticity, sometimes called *intrinsic attenuation*, involves conversion of seismic energy to heat.

As in many seismological applications, it is worth first considering familiar analogous behaviors for light. As you move away from a street lamp at night, the light appears dimmer for several reasons. The first is geometric spreading: light moves outward from the lamp in expanding spherical wave fronts (Section 2.4.3). By the conservation of energy, the energy in a unit area of the growing wave front decreases as r^{-2} , where r is the radius of the sphere or distance from the lamp.

Second, the light dims as it is scattered by air molecules, dust, and water in the air. As we have discussed, scattering results when objects acting as Huygens' sources scatter energy in all directions. This effect is dramatic on a foggy night because the scattered light causes a halo around the lamp.

Third, the light is focused or defocused by changes in the refractive properties of the air.¹ This effect is termed *multipathing* in seismology. Focusing and defocusing can be illustrated by looking at the street light through binoculars. Looking through binoculars the usual way, the waves are focused by the lenses, and the lamp appears closer and brighter. Reversing the binoculars makes the lamp appear further away and dimmer.

Fourth, some of the light energy is absorbed by the air and converted to heat. This process differs from the other three in that light energy is actually lost, not just moved onto a different path.

All four processes are important for seismic waves. The first three are described by elastic wave theory, and can increase or decrease an arrival's amplitude by shifting energy within the wave field. By contrast, anelasticity reduces wave amplitudes only because energy is lost from the elastic waves. So much of seismology is built upon the approximation that the earth responds elastically during seismic propagation that it is easy to forget that the earth is not perfectly elastic. However, without anelasticity, seismic waves from every earthquake that ever occurred would still be reverberating until the accumulating reverberations shattered the earth. Elasticity is a good approximation for the earth's response to seismic waves, but

¹ This process causes mirages, where light is refracted differently by hot air just above the ground. Similarly the distorted appearance of the setting sun results from seeing different parts of it through different levels of the atmosphere which refract light differently because of the vertical density gradient.